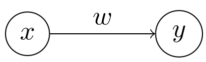
1 Twice example

1.1 Model



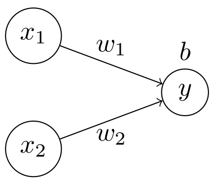
$$y = x \cdot w$$

1.2 Cost

$$\begin{aligned}
C &= \frac{1}{n}\sum_{i=1}^{n}\left(x_i w - y_i\right)^2 \\
\frac{dC}{dw} &= \frac{d}{dw}\left(\frac{1}{n}\sum_{i=1}^{n}\left(x_i w - y_i\right)^2\right) \\
&= \frac{1}{n}\frac{d}{dw}\left(\sum_{i=1}^{n}\left(x_i w - y_i\right)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\frac{d}{dw}\left(\left(x_i w - y_i\right)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}2\left(x_i w - y_i\right)x_i
\end{aligned}$$

2 Gates example

2.1 Model



$$\begin{aligned}
&y = \sigma(x_1 w_1 + x_2 w_2 + b) \\
&\sigma(x) = \frac{1}{1+e^{-x}} \\
&\frac{d}{dx}\sigma(x) = \sigma(x)(1-\sigma(x))
\end{aligned}$$

2.2 Cost

$$
\begin{aligned}
a_i &= \sigma(x_{1i}w_1 + x_{2i}w_2 + b) \\
\frac{\partial a_i}{\partial w_1} &= \sigma(x_{1i}w_1 + x_{2i}w_2 + b)(1 - \sigma(x_{1i}w_1 + x_{2i}w_2 + b))x_{1i} \\
&= a_i(1 - a_i)x_{1i} \\
\frac{\partial a_i}{\partial w_2} &= \sigma(x_{1i}w_1 + x_{2i}w_2 + b)(1 - \sigma(x_{1i}w_1 + x_{2i}w_2 + b))x_{2i} \\
&= a_i(1 - a_i)x_{2i} \\
\frac{\partial a_i}{\partial b} &= \sigma(x_{1i}w_1 + x_{2i}w_2 + b)(1 - \sigma(x_{1i}w_1 + x_{2i}w_2 + b)) \\
&= a_i(1 - a_i) \\
C &= \frac{1}{n}\sum_{i=1}^{n}(a_i - y_i)^2 \\
\frac{\partial C}{\partial w_1} &= \frac{\partial}{\partial w_1}\left(\frac{1}{n}\sum_{i=1}^{n}(a_i - y_i)^2\right) \\
&= \frac{1}{n}\frac{\partial}{\partial w_1}\left(\sum_{i=1}^{n}(a_i - y_i)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial w_1}\left((a_i - y_i)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}2(a_i - y_i)a_i(1 - a_i)x_{1i} \\
\frac{\partial C}{\partial w_2} &= \frac{\partial}{\partial w_1}\left(\frac{1}{n}\sum_{i=1}^{n}(a_i - y_i)^2\right) \\
&= \frac{1}{n}\frac{\partial}{\partial w_1}\left(\sum_{i=1}^{n}(a_i - y_i)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial w_1}\left((a_i - y_i)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}2(a_i - y_i)a_i(1 - a_i)x_{2i} \\
\frac{\partial C}{\partial b} &= \frac{\partial}{\partial w_1}\left(\frac{1}{n}\sum_{i=1}^{n}(a_i - y_i)^2\right) \\
&= \frac{1}{n}\frac{\partial}{\partial w_1}\left(\sum_{i=1}^{n}(a_i - y_i)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial w_1}\left((a_i - y_i)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}2(a_i - y_i)a_i(1 - a_i)
\end{aligned}
$$

2.3 Cost - Simplified

$$
\begin{aligned}
a_i &= \sigma(x_{1i}w_1 + x_{2i}w_2 + b) \\
\frac{\partial a_i}{\partial w_j} &= a_i(1 - a_i)x_{ji} \\
\frac{\partial a_i}{\partial b} &= a_i(1 - a_i) \\
C &= \frac{1}{n}\sum_{i=1}^{n}(a_i - y_i)^2 \\
\frac{\partial C}{\partial w_j} &= \frac{1}{n}\sum_{i=1}^{n}2(a_i - y_i)a_i(1 - a_i)x_{ji} \\
\frac{\partial C}{\partial b} &= \frac{1}{n}\sum_{i=1}^{n}2(a_i - y_i)a_i(1 - a_i)
\end{aligned}
$$